DS ASSIGNMENT 1 -KARISHUMA VIKRAM

1.

```
#include <stdio.h>

int main()
{
   int r1, c1, r2, c2;
   int A[20][20], B[20][20], C[20][20];

   printf("Enter the rows and columns of the first matrix: ");
   scanf("%d %d", &r1, &c1);

   printf("Enter the rows and columns of the second matrix: ");
   scanf("%d %d", &r2, &c2);

   if (r1 != r2 || c1 != c2)
   {
      printf("Matrix addition is not possible.\n");
      return 0;
   }

   printf("Enter the elements of the first matrix:\n");
   for (int i = 0; i < r1; i++)
   {
      for (int j = 0; j < c1; j++)
      {
         scanf("%d", &A[i][j]);
      }
   }

   printf("Enter the elements of the second matrix:\n");
```

```
    for (int i = 0; i < r2; i++)
    {
        for (int j = 0; j < c2; j++)
        {
            scanf("%d", &B[i][j]);
        }
    }

    for (int i = 0; i < r1; i++)
    {
        for (int j = 0; j < c1; j++)
        {
            C[i][j] = A[i][j] + B[i][j];
        }
    }

    printf("\nMatrix Addition Result:\n");
    for (int i = 0; i < r1; i++)
    {
        for (int j = 0; j < c1; j++)
        {
            printf("%d ", C[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter the rows and columns of the first matrix: 2 2
Enter the rows and columns of the second matrix: 2 2
Enter the elements of the first matrix:
1 3 5 7
Enter the elements of the second matrix:
2 4 6 8

Matrix Addition Result:
3 7
11 15


=== Code Execution Successful ===
```

2. Write a C program to read n numbers into an array and display them in reverse order

Program:

```
#include <stdio.h>
int main()
{
int n,arr[20],reverse[20],j=0;
printf("enter the number of elements in the array\n");
scanf("%d",& n);
printf("enter the elements in the array\n");
for(int i=0;i<n;i++)
{
scanf("%d",&arr[i]);
   }
   printf("the elements in the array are:\n");
   for(int i=0;i<n;i++)
   {
      printf("%d\t",arr[i]);
   }
   printf("\n");
   for(int i=n-1;i>=0;i--)
    {
```

```
      reverse[j]=arr[i];
      j++;
    }
    printf("reverse of the elements in the array\n");
  for(int i=0;i<n;i++)
  {
      printf("%d\t",reverse[i]);
  }
 return 0;
}
```

Output

```
enter the number of elements in the array
5
enter the elements in the array
1 2 3 4 5
the elements in the array are:
1   2   3   4   5
reverse of the elements in the array
5   4   3   2   1

=== Code Execution Successful ===
```

3. Count the total number of duplicate elements in an array.

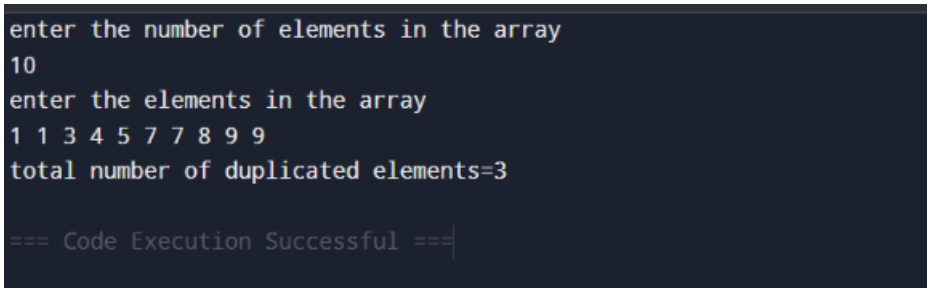
Program

```
#include <stdio.h>
int main()
{
  int arr[20],n;
  printf("enter the number of elements in the array\n");
  scanf("%d",& n);
  printf("enter the elements in the array\n");
  for(int i=0;i<n;i++)
  {
     scanf("%d",&arr[i]);
```

```
    }
    int count=0;
    for(int j=0;j<n;j++)
    {
        int dup=0;
        int a=0;
        for(int k=0;k<j;k++)
        {
            if(arr[k]==arr[j])
            {
                a=1;
                break;
            }
        }
        if(a)
        {
            continue;
        }
    for(int i=j+1;i<n;i++)
    {
        if(arr[i]==arr[j])
        {
        dup=1;
        break;
        }
    }
    if(dup)
    {
        count++;
    }
}
```

```
printf("total number of duplicated elements=%d",count);
return 0;
}
```

Output



4. Print unique elements in an array.

Program

```
# include <stdio.h>
int main()
{
   int arr[20],n;
   printf("enter the number of elements in the array\n");
   scanf("%d",& n);
   printf("enter the elements in the array\n");
   for(int i=0;i<n;i++)
   {
      scanf("%d",&arr[i]);
   }
   int count=0;
   int unique=0;
   for(int j=0;j<n;j++)
   {
      int dup=0;
      int a=0;
      for(int k=0;k<j;k++)
      {
```

```
            if(arr[k]==arr[j])
            {
               a=1;
               break;
            }
         }
         if(a)
         {
            continue;
         }
      for(int i=j+1;i<n;i++)
      {
         if(arr[i]==arr[j])
         {
         dup=1;
         break;
         }
      }
      if(dup)
      {
         count++;
      }
      else
      {
         unique++;
      }
}
printf("total number of unique elements=%d",unique);
return 0;
}
```

Output

```
enter the number of elements in the array
5
enter the elements in the array
1 9 9 3 4
total number of unique elements=3

=== Code Execution Successful ===
```

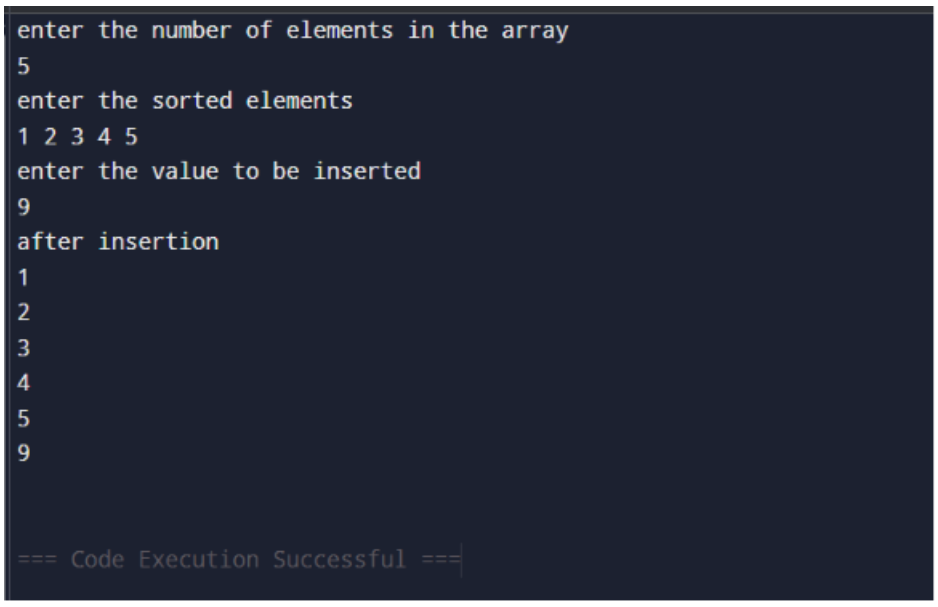
5. Insert an element in a sorted array.

Program

```
# include <stdio.h>
int main()
{
   int arr[20],n,val,ins=0;
   printf("enter the number of elements in the array\n");
   scanf("%d",&n);
   printf("enter the sorted elements\n");
   for(int i=0;i<n;i++)
   {
   scanf("%d",&arr[i]);
   }
   printf("enter the value to be inserted\n");
   scanf("%d",&val);
   for(int i=0;i<n;i++)
   {
      if(val<arr[i])
      {
         for(int j=n;j>i;j--)
         {
         arr[j]=arr[j-1];
         }
         arr[i]=val;
         ins=1;
```

```
        break;
      }
   }
   if(ins==0)
   {
arr[n]=val;
}
printf("after insertion\n");
for(int i=0;i<=n;i++)
{
   printf("%d\n",arr[i]);
}
}
```

Output



6. Insert an element in an unsorted array.

Program

```
# include <stdio.h>
int main()
{
int arr[20],n,val,pos;
printf("enter the number of elements in the array\n");
```

```
    scanf("%d",&n);
    printf("enter the  elements\n");
    for(int i=0;i<n;i++)
    {
    scanf("%d",&arr[i]);
    }
    printf("enter the value to be inserted\n");
    scanf("%d",&val);
    printf("enter the position needed to insert\n");
    scanf("%d",&pos);
    for(int i=n;i>pos;i--)
    {
       arr[i]=arr[i-1];
    }
    arr[pos]=val;
    printf("after insertion\n");
    for(int i=0;i<=n;i++)
    {
       printf("%d\n",arr[i]);
    }
}
```

Output

```
enter the number of elements in the array
5
enter the  elements
1 2 3 4 5
enter the value to be inserted
7
enter the position needed to insert
2
after insertion
1
2
7
3
4
5

=== Code Execution Successful ===
```

7. Matrix multiplication

Program

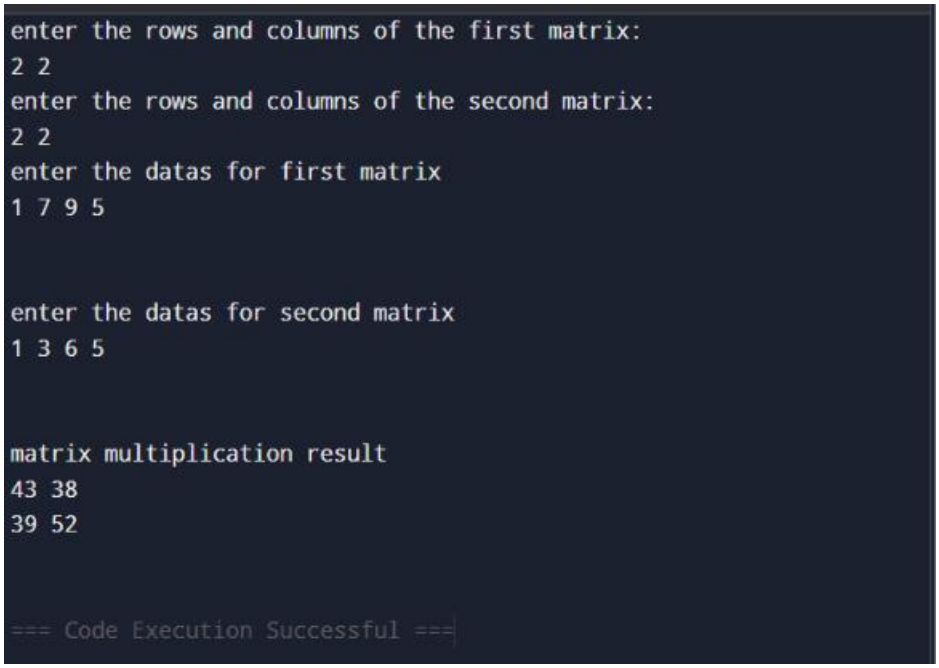
```
# include <stdio.h>
int main()
{
int r1,r2,c1,c2,A[20][20],B[20][20],C[20][20];
printf("enter the rows and columns of the first matrix:\n");
scanf("%d %d",&r1,&c1);
printf("enter the rows and columns of the second matrix:\n");
scanf("%d %d",&r2,&c2);
if(c1!=r2)
{
printf("invalid,matrix multiplication can't be performed");
return 0;
}
printf("enter the datas for first matrix\n");
  for(int i=0;i<r1;i++)
{
  for(int j=0;j<c1;j++)
```

```
{
    scanf("%d",& A[i][j]);
}
printf("\n");
}
printf("enter the datas for second matrix\n");
for(int i=0;i<r2;i++)
{
 for(int j=0;j<c2;j++)
{
   scanf("%d",& B[i][j]);
}
printf("\n");
}
for(int i=0;i<r1;i++)
{
 for(int j=0;j<c2;j++)
{
   C[i][j]=0;
   for(int k=0;k<r2;k++)
   {
    C[i][j]+=A[i][k]*B[k][j];
   }
}
}
printf("matrix multiplication result\n");
for(int i=0;i<r1;i++)
{

for(int j=0;j<c2;j++)
{
```

```
printf("%d ",C[i][j]);
}
printf("\n"); }
return 0;
}
```

Output



8. Matrix transpose.

Program

```
# include <stdio.h>
int main()
{
int r1,r2,c1,c2,A[20][20];
printf("enter the rows and columns of the first matrix:\n");
scanf("%d %d",&r1,&c1);
printf("enter the datas for matrix\n");
for(int i=0;i<r1;i++)
{

for(int j=0;j<c1;j++)
```

```
{
scanf("%d",& A[i][j]);
}
printf("\n");
}
printf("transpose of the matrix\n");
for(int i=0;i<c1;i++)
{
{
}
for(int j=0;j<r1;j++)
printf("%d ",A[j][i]);
printf("\n");
}
return 0;
}
```

Output

```
enter the rows and columns of the first matrix:
2  2
enter the datas for matrix
1 2 3 4


transpose of the matrix
1 3
2 4


=== Code Execution Successful ===
```

9. Right diagonal sum of a matrix

Program

```
# include <stdio.h>
int main()
{
```

```
    int n,A[10][10],sum=0;
    printf("enter the order of the square matrix\n");
    scanf("%d",&n);
    printf("enter the values of the matrix\n");
    for(int i=0;i<n;i++)
    {
        for(int j=0;j<n;j++)
        {
            scanf("%d",&A[i][j]);
        }
        printf("\n");
    }
for(int i=0;i<n;i++)
{
sum=sum+A[i][n-i-1];
}
printf("Right diagonal sum=%d",sum);
return 0;
}
```

Output

```
enter the order of the square matrix
2
enter the values of the matrix
1 4 5 6


Right diagonal sum=9

=== Code Execution Successful ===
```

10. Check whether a matrix is an identity matrix.

Program

```
# include <stdio.h>
int main()
```

```
{
int n,A[10][10],f=1;
printf("enter the order of the square matrix\n");
scanf("%d",&n);
printf("enter the values of the matrix\n");
for(int i=0;i<n;i++)
{
      for(int j=0;j<n;j++)
      {
         scanf("%d",&A[i][j]);
      }
      printf("\n");
   }
   for(int i=0;i<n;i++)
   {
      for(int j=0;j<n;j++)
      {
        if(i==j)
        {
         if(A[i][j]!=1)
         {
            f=0;
            break;
         }
        }
        else
        {
         if(A[i][j]!=0)
         {
            f=0;
            break;
```

```
        }
       }
      }
      if(f==0)
      break;
   }
   if(f==1)
   {
      printf("the given matrix is an identity matrix\n");
   }
   else{
      printf("the given matrix is not an identity matrix\n");
   }
   return 0;
}
```

Output

```
enter the order of the square matrix
2
enter the values of the matrix
1 0 0 1


the given matrix is an identity matrix


=== Code Execution Successful ===
```

11. Find the number of vowels, consonants, digits, and white spaces in a string.

Program

```
# include <stdio.h>
int main()
{
   char str[50];
   printf("enter the string\n");
   fgets(str,50,stdin);
```

```
   int i=0,vow=0,con=0,space=0,num=0;
   while(str[i]!='\0')
   {
      if(('a'<=str[i] && 'z'>=str[i])||('A'<=str[i]&& 'Z'>=str[i]))
      {
      if(str[i]=='A'||str[i]=='E'||str[i]=='I'||str[i]=='O'||str[i]=='U'||str[i]=='a'||str[i]=='i'||str[i
]=='e'||str[i]=='o'||str[i]=='u')
      {
          vow++;
      }

   else
   {
      con++;
   }
}
if(str[i]==' ')
{
   space++;
}
if(str[i]>='0' && str[i]<='9')
{
   num++;
}
i++;
   }
   printf("number of vowels in the string=%d\n",vow);
   printf("number of consonents in the string=%d\n",con);
   printf("number of spaces in the string=%d\n",space);
   printf("number present in the string count =%d\n",num);
return 0;
```

}

Output

```
enter the string
my name is karishuma
number of vowels in the string=7
number of consonents in the string=10
number of spaces in the string=3
number present in the string count =0


=== Code Execution Successful ===
```

12. Find the frequency of a character in a string.

Program

```
#include <stdio.h>
int main() {
    char str[100];
    int i, j, count;

    printf("Enter string: ");
    scanf("%s", str);

    for (i = 0; str[i] != '\0'; i++) {

        if (str[i] == '*')
            continue;

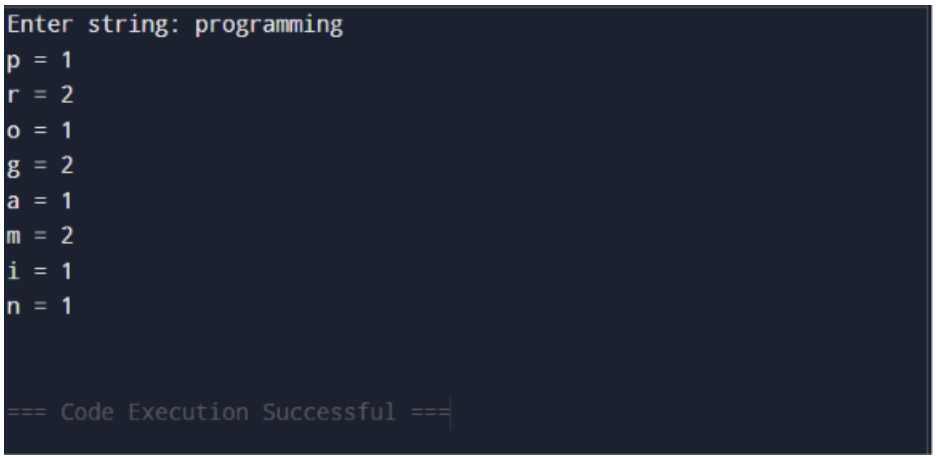
        count = 1;

        for (j = i + 1; str[j] != '\0'; j++) {
            if (str[i] == str[j]) {
                count++;
                str[j] = '*';
```

```
        }
      }

      printf("%c = %d\n", str[i], count);
   }
   return 0;
}
```

Output



13. Count the number of words in a string.

Program

```
# include <stdio.h>
int main()
{
   char str[50];
   printf("enter the string\n");
   fgets(str,50,stdin);
   int i=0,space=0;
   while(str[i]!='\0')
   {
      if(str[i]==' ')
      {
         space++;
      }
```

```
    i++;
  }
  printf("count of words in the string=%d",space+1);
  return 0;
}
```

Output

```
enter the string
my name is karishuma vikram
count of words in the string=5

=== Code Execution Successful ===
```

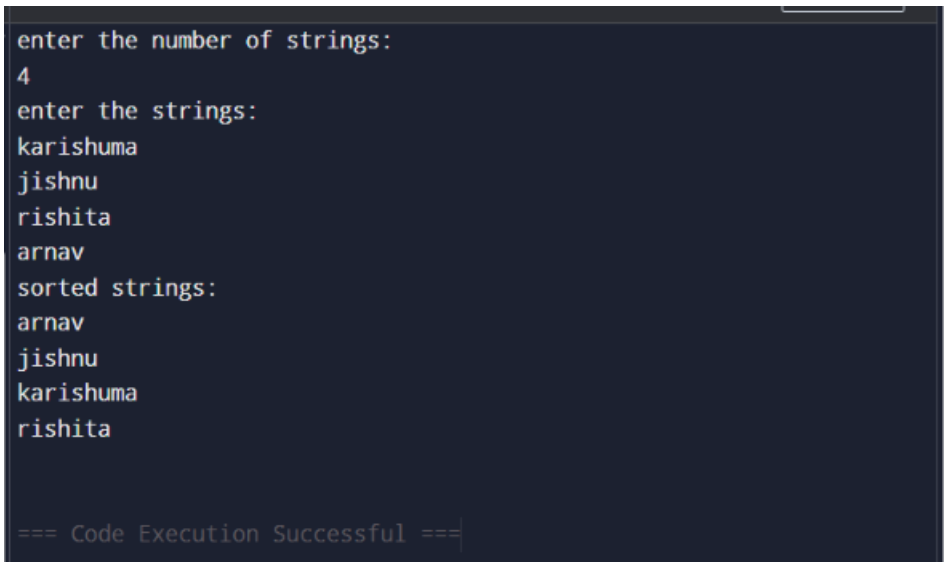
14. Sort a string array.

Program

```
# include <stdio.h>
# include <string.h>
int main()
{
  int n;
  char str[10][30],temp[30];
  printf("enter the number of strings:\n");
  scanf("%d",&n);
  getchar();
  printf("enter the strings:\n");
  for(int i=0;i<n;i++)
  {
    fgets(str[i],30,stdin);
    str[i][strcspn(str[i],"\n")]='\0';
  }
  for(int i=0;i<n-1;i++)
  {
```

```
    for(int j=i+1;j<n;j++)
    {
       if(strcmp(str[i],str[j])>0)
       {
         strcpy(temp,str[i]);
         strcpy(str[i],str[j]);
         strcpy(str[j],temp);
       }
    }
  }
  printf("sorted strings:\n");
  for(int i=0;i<n;i++)
  {
    printf("%s\n",str[i]);
  }
return 0;
}
```

Output



15. Toggle the case of a sentence (lowercase to uppercase and vice versa

Program

```
# include <stdio.h>
```

```
int main()
{
   char str[100];
   printf("enter a sentence:\n");
   fgets(str,100,stdin);
   for(int i=0;str[i]!='\0';i++)
   {
      if(str[i]>='a' && str[i]<='z')
      {
         str[i]=str[i]-32;
      }
     else if(str[i]>='A' && str[i]<='Z')
      {
         str[i]=str[i]+32;
      }
   }
   printf("toggled sentence:\n");
printf("%s",str);
return 0;
}
```

Output

